P02

Transient Response of Series RLC Resonant Circuits

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Domain DOM_ENG

Abstract

An electrical series RLC circuit excited by step or impulse voltage exhibits transient dynamics governed by identical differential equations. Inductance models inertia, capacitance models compliance, and resistance models dissipation.

Abstract quoted from C. K. Alexander et al., Fundamentals of Electric Circuits (2000).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?